

POTCHEFSTROOM, South Africa

WMO: 683500

Lat: 26.7358S

Lon: 27.0756E

Elev: 4432

StdP: 12.49

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 28.8	(c) 31.7	(d) -1.0	(e) 6.1	(f) 47.3	(g) 6.1	(h) 8.8	(i) 54.6	(j) 19.7	(k) 70.0	(l) 17.4	(m) 67.0	(n) 2.7	(o) 20	(p) 0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 22.9	(c) 93.6	(d) 62.8	(e) 91.1	(f) 62.8	(g) 89.0	(h) 62.7	(i) 70.8	(j) 80.6	(k) 69.3	(l) 80.7	(m) 68.2	(n) 80.3	(o) 7.9	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	122.5	74.6	66.3	114.3	72.7	64.8	108.6	71.8	37.8	80.3	36.4	80.9	35.3	80.0	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.8	(b) 16.3	(c) 14.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 18.8	(b) 16.3	(c) 14.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	73.7	72.2	70.1	63.8	57.3	51.9	51.4	57.2	64.1	69.1	70.8	72.9
	DBStd	9.37	3.93	3.66	4.19	4.73	4.53	4.97	5.48	6.64	6.38	5.90	4.87	4.39
	HDD50	110	0	0	0	1	7	34	49	17	2	0	0	0
	HDD65	1524	1	2	9	75	239	392	423	251	91	29	9	3
	CDD50	5403	734	622	622	415	234	92	91	239	425	594	625	710
	CDD65	1342	270	204	166	39	1	0	0	9	64	157	184	248
	CDH74	14310	2490	1729	1528	553	108	18	22	331	1226	1982	1976	2347
	CDH80	5343	1038	635	516	92	7	0	0	50	411	822	796	976
(14) Wind	WSAvg	5.9	5.9	5.3	4.8	4.6	4.6	4.8	4.9	6.6	7.1	7.4	7.4	7.2
(15) Precipitation	PrecAvg	23.9	4.2	3.5	2.8	2.1	0.6	0.2	0.1	0.4	0.7	2.0	3.2	3.8
	PrecMax	42.3	10.0	9.4	8.2	6.3	4.3	1.7	0.9	3.2	4.6	5.6	6.5	8.5
	PrecMin	8.8	0.0	0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	1.8
	PrecStd	6.5	2.1	2.1	1.5	1.7	0.9	0.4	0.2	0.6	1.0	1.4	1.7	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.0	92.6	92.3	86.8	81.3	76.8	77.3	84.7	90.3	95.4	95.8	96.6
		MCWB	64.5	63.1	64.1	62.1	58.4	54.3	52.5	54.8	57.4	61.0	61.9	62.6
	2%	DB	92.3	89.6	88.6	82.7	77.0	72.5	73.3	80.9	87.7	91.3	91.7	91.8
		MCWB	65.1	63.3	64.0	60.9	56.7	52.2	50.6	53.4	57.5	60.3	61.3	63.7
	5%	DB	89.3	87.1	85.6	80.1	74.7	69.8	70.2	77.6	85.1	88.1	88.1	88.8
		MCWB	65.8	64.5	64.0	60.1	55.8	50.9	49.7	52.6	56.6	59.9	61.9	63.8
	10%	DB	86.3	84.2	82.7	77.4	72.1	67.0	67.3	73.9	81.9	84.8	85.0	85.6
		MCWB	65.6	65.0	64.1	60.0	55.0	49.8	49.0	51.6	55.7	59.8	62.3	64.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.9	71.2	70.1	66.6	62.2	58.4	56.7	60.4	63.3	75.0	73.4	72.5
		MCDB	85.4	80.9	81.0	76.3	71.7	69.1	69.9	72.8	81.0	75.2	79.6	80.4
	2%	WB	69.8	69.4	68.6	64.7	59.5	55.6	53.9	57.0	61.1	67.5	69.0	69.7
		MCDB	83.7	80.6	80.8	75.9	72.4	66.9	67.4	73.5	79.5	79.5	78.5	79.4
	5%	WB	68.4	68.4	67.3	63.4	58.0	53.2	52.0	55.2	59.1	64.6	66.9	68.1
		MCDB	81.8	80.2	79.7	74.9	71.1	65.1	66.8	72.4	77.2	78.8	78.2	79.7
	10%	WB	67.4	67.4	66.1	62.0	56.3	51.3	50.0	53.4	57.4	62.6	65.3	66.7
		MCDB	80.4	78.9	77.8	73.3	68.7	64.2	65.2	71.5	76.3	77.1	77.8	79.6
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	22.7	23.5	25.6	29.7	31.2	31.7	32.3	31.7	27.7	25.5	23.6
		MCDBR	28.7	28.4	28.9	29.8	33.5	34.0	34.3	34.9	35.8	32.8	31.0	29.1
		MCWBR	9.5	10.0	9.9	11.7	15.0	16.9	16.8	16.0	14.5	12.4	10.3	10.2
	5% WB	MCDBR	22.7	21.5	22.4	23.5	27.1	28.7	29.6	30.1	29.3	25.2	22.1	22.1
		MCWBR	8.9	8.6	9.4	11.1	14.1	16.4	16.1	14.9	12.4	12.7	10.5	9.4
(40) Clear-Sky Solar Irradiance	taub		0.382	0.387	0.372	0.326	0.290	0.274	0.274	0.328	0.371	0.400	0.354	0.364
	taud		2.341	2.329	2.349	2.447	2.521	2.533	2.510	2.354	2.244	2.204	2.380	2.382
	Ebn at Noon		305	298	293	292	291	290	294	287	288	291	312	311
	Edh at Noon		43	42	39	32	27	25	27	35	43	48	41	41
(44) All-Sky Solar Radiation	RadAvg		2169	2013	1789	1518	1363	1232	1342	1599	1932	2085	2222	2253
(45)	RadStd		224	181	148	124	62	54	37	79	82	140	173	132

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page